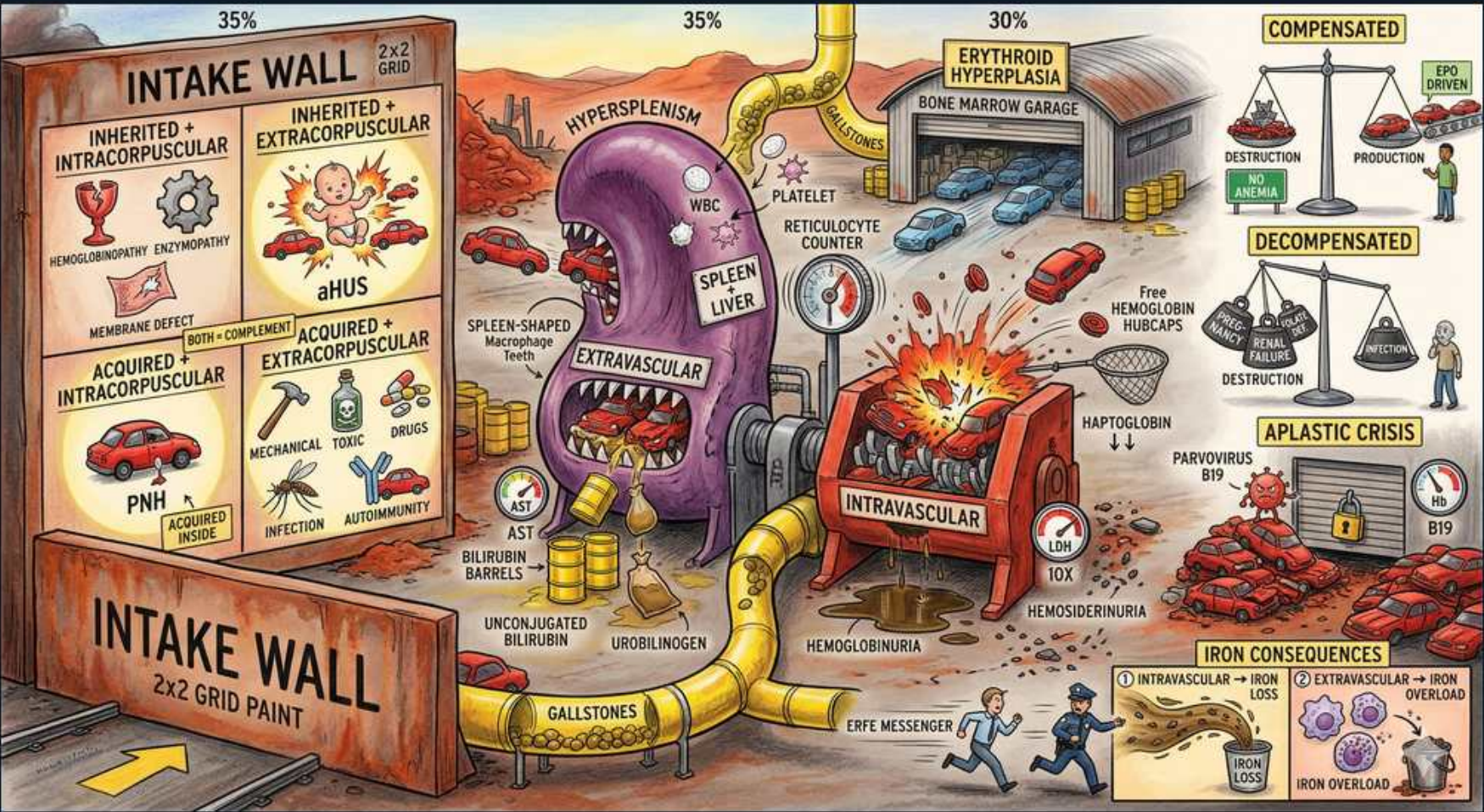


Hemolytic Anemia: Intravascular vs Extravascular



1. Intake wall 2x2 grid

WHAT YOU SEE

Wall — inherited/acquired x intra/extracorporeal

WHAT IT ENCODES

Classify hemolysis on two axes: inherited vs acquired, and intracorporeal (defect inside RBC) vs extracorporeal (defect outside). Inherited+intracorporeal = membrane/enzyme/Hb defects; acquired+extracorporeal = immune, MAHA, infection. This 2x2 organizes the whole differential.

BOARD TRAP

Mixing axes — e.g. calling PNH inherited; PNH is an ACQUIRED intracorporeal defect (somatic PIGA mutation).

2. PNH = acquired intracorporeal

WHAT YOU SEE

Boxed PNH, acquired inside

WHAT IT ENCODES

PNH is the exception: an ACQUIRED (somatic PIGA mutation) yet INTRACORPUSCULAR defect, lacking GPI-anchored CD55/CD59 so complement lyses RBCs. It is the one acquired cause that is intrinsic to the cell. Causes intravascular hemolysis and thrombosis.

BOARD TRAP

Classifying PNH as inherited because the defect is intrinsic; it is acquired (clonal somatic mutation).

3. Hereditary spherocytosis

WHAT YOU SEE

Inherited intracorporeal — membrane defect

WHAT IT ENCODES

Inherited intracorporeal causes: membrane (hereditary spherocytosis — ankyrin/spectrin, EMA-binding test, splenectomy curative), enzyme (G6PD, PK), and hemoglobinopathy (sickle, thalassemia). These RBCs are destroyed extravascularly in the spleen.

BOARD TRAP

Ordering osmotic fragility as definitive for HS; EMA binding flow cytometry is now the preferred confirmatory test.

4. Hypersplenism / extravascular

WHAT YOU SEE

Spleen monster with teeth

WHAT IT ENCODES

Extravascular hemolysis: splenic macrophages remove antibody- or defect-coated RBCs. Markers — high UNCONJUGATED bilirubin, high urobilinogen, splenomegaly, spherocytes, pigment gallstones. Haptoglobin only mildly low. Warm AIHA and HS are classic.

BOARD TRAP

Expecting hemoglobinuria in extravascular hemolysis; free Hb stays low, so urine hemoglobin is an INTRAVascular marker.

5. Unconjugated bilirubin

WHAT YOU SEE

Bilirubin barrels

WHAT IT ENCODES

Hemolysis releases heme, raising INDIRECT (unconjugated) bilirubin and urobilinogen — jaundice and pigment (calcium bilirubinate) gallstones. LDH rises and haptoglobin falls. These confirm hemolysis regardless of site.

BOARD TRAP

Seeing conjugated hyperbilirubinemia and calling it hemolysis; hemolytic jaundice is unconjugated/indirect.

6. Intravascular hemolysis

WHAT YOU SEE

Explosion — RBCs lysing in vessel

WHAT IT ENCODES

Intravascular hemolysis lyses RBCs in the circulation: very low/absent haptoglobin, high LDH, hemoglobinemia, hemoglobinuria, hemosiderinuria. Causes — PNH, MAHA (TTP/HUS/DIC), transfusion reactions, mechanical valves, severe G6PD crisis.

BOARD TRAP

Expecting prominent splenomegaly in intravascular hemolysis; destruction is in vessels, spleen is not the main site.

7. Haptoglobin

WHAT YOU SEE

Free hemoglobin hubcaps

WHAT IT ENCODES

Free plasma Hb binds haptoglobin, which is then cleared — so haptoglobin DROPS, most markedly in intravascular hemolysis. A low haptoglobin with high LDH and reticulocytosis is the sensitive signature of hemolysis.

BOARD TRAP

Using a normal haptoglobin to exclude hemolysis; it is an acute-phase reactant and can be falsely normal with inflammation.

8. Hemoglobinuria

WHAT YOU SEE

Red urine, hemosiderinuria

WHAT IT ENCODES

When haptoglobin is saturated, free Hb is filtered: hemoglobinuria (early) and hemosiderinuria (days later, from tubular cell turnover). These are specific for INTRAVascular hemolysis. Dipstick positive for blood but no RBCs on micro.

BOARD TRAP

Calling hematuria when dipstick is blood-positive but no RBCs are seen; that pattern is hemoglobinuria/myoglobinuria.

9. Erythroid hyperplasia

WHAT YOU SEE

Bone marrow garage, EPO driven

WHAT IT ENCODES

Marrow compensates with erythroid hyperplasia driven by EPO, raising the reticulocyte count and reticulocyte production index. A retic count that fails to rise signals an aplastic crisis or coexisting marrow problem.

BOARD TRAP

Expecting a low retic count in hemolysis; reticulocytosis is the hallmark unless marrow fails (aplastic crisis).

10. Reticulocyte counter

WHAT YOU SEE

Retic counter gauge

WHAT IT ENCODES

Reticulocytosis confirms an appropriate marrow response to hemolysis. Use the reticulocyte production index (corrected for anemia/shift) to judge adequacy. Hemolytic anemia is a hyperproliferative (high-retic) anemia.

BOARD TRAP

Confusing high retics from hemolysis with high retics from acute blood loss; both raise retics — use clinical context and hemolysis labs.

11. Compensated vs decompensated

WHAT YOU SEE

Scales — production vs destruction

WHAT IT ENCODES

Compensated hemolysis: marrow production matches destruction, hemoglobin normal despite high turnover (high LDH/retics, low haptoglobin, no anemia). Decompensated: destruction outpaces production and anemia appears. Severity depends on this balance.

BOARD TRAP

Excluding hemolysis because hemoglobin is normal; compensated hemolysis has normal Hb with abnormal hemolysis labs.

12. Aplastic crisis

WHAT YOU SEE

Parvovirus B19 lock

WHAT IT ENCODES

Parvovirus B19 infects erythroid progenitors and halts production for ~7-10 days. In chronic hemolysis (HS, sickle) this causes a sudden APLASTIC CRISIS — abrupt Hb drop with a LOW reticulocyte count. Supportive transfusion; self-limited.

BOARD TRAP

Seeing a low retic in a known hemolytic patient and missing parvovirus B19 aplastic crisis.

13. Iron consequences

WHAT YOU SEE

Intravascular iron loss vs extravascular overload

WHAT IT ENCODES

INTRAVascular hemolysis loses iron in the urine (hemosiderinuria) and can cause iron DEFICIENCY (e.g. chronic PNH). EXTRAvascular hemolysis recycles iron internally and, with transfusions, leads to iron OVERLOAD. Site predicts iron balance.

BOARD TRAP

Assuming all chronic hemolysis causes iron overload; chronic intravascular hemolysis can deplete iron via urinary loss.

14. Schistocytes / MAHA

WHAT YOU SEE

Fragmented RBCs near the machine

WHAT IT ENCODES

Mechanical/microangiopathic hemolysis shears RBCs into schistocytes — seen in thrombotic microangiopathies, DIC, malignant hypertension, prosthetic valves. Schistocytes plus low platelets demand urgent workup. This is intravascular and often life-threatening.

BOARD TRAP

Dismissing a few schistocytes; with thrombocytopenia they signal microangiopathy needing emergent evaluation.

15. WBC / platelet bystanders

WHAT YOU SEE

WBC and platelet cars

WHAT IT ENCODES

Pure hemolysis spares WBCs and platelets. If anemia is accompanied by cytopenias (low WBC/platelets), think marrow infiltration, PNH, TMA, or hypersplenism rather than isolated hemolysis. Pattern of cell lines narrows the cause.

BOARD TRAP

Calling isolated hemolysis when pancytopenia is present; multilineage cytopenia points to PNH, TMA, or marrow disease.

16. Direct antiglobulin (Coombs)

WHAT YOU SEE

Autoimmunity / drugs panel

WHAT IT ENCODES

The direct antiglobulin (Coombs) test separates immune (DAT+) from non-immune (DAT-) acquired hemolysis. DAT+ = warm/cold AIHA, drug-induced; DAT- with schistocytes = MAHA. It is the first branch point in acquired hemolysis.

BOARD TRAP

Skipping the DAT in acquired hemolysis; it is the single test that splits immune from microangiopathic causes.